

Experiment No. 9

Title:

Demonstration of Firewall Configuration and Traffic Control Using GFW/UFW in Kali Linux and Ubuntu Environment

Aim:

To demonstrate the installation, configuration, and implementation of firewall rules using **GFW/UFW**, and to analyse their impact on network communication in the Online Examination System environment.

Objectives:

After completing this experiment, students will be able to:

1. Understand the concept of host-based firewall.
2. Configure firewall rules using CLI and GUI.
3. Apply inbound and outbound filtering policies.
4. Block and allow specific ports.
5. Verify firewall rule impact on client-server communication.
6. Integrate firewall protection with secure applications.

Theory:

What is a Firewall?

A firewall is a network security system that monitors and controls incoming and outgoing traffic based on predefined security rules.

It acts as a barrier between:

- Trusted internal network
- Untrusted external network

Types of Firewalls

1. Packet Filtering Firewall
2. Stateful Inspection Firewall
3. Application-Level Firewall
4. Next-Generation Firewall

In this experiment, we use a **host-based packet filtering firewall**.

UFW (Uncomplicated Firewall)

UFW

UFW is a command-line firewall tool used in Linux to manage iptables easily.

It provides:

- Simple syntax
- Default deny/allow policies
- Port-based filtering
- IP-based filtering

GFW

GFW

GFW is a graphical interface for UFW.

It allows:

- Easy rule creation
- Enable/disable firewall
- Monitor traffic rules visually

Role in Online Examination System

In your capstone project:

- Ubuntu → Exam Server (Port 5000)
- Keycloak → Authentication (Port 8080)
- Kali → Client

Firewall ensures:

- Only authorized ports are accessible
- Prevents unauthorized access
- Protects exam system from attacks

Tools Required

- Ubuntu (Server VM)
- Kali Linux (Client VM)
- Java Exam Server & Client
- UFW
- GFW

Procedure:

Step 1: Install UFW (if not installed)

```
sudo apt update
```

```
sudo apt install ufw -y
```

Step 2: Install GFW (Optional GUI)

```
sudo apt install gufw -y
```

Launch GUI:

```
gufw
```

Step 3: Enable Firewall

```
sudo ufw enable
```

Check status:

```
sudo ufw status verbose
```

Expected default output:

Default: deny (incoming)

Default: allow (outgoing)

Step 4: Configure Default Policies

```
sudo ufw default deny incoming
```

```
sudo ufw default allow outgoing
```

Step 5: Allow Required Ports

Allow Keycloak authentication server:

```
sudo ufw allow 8080
```

Allow Exam Server socket port:

```
sudo ufw allow 5000
```

Verify:

```
sudo ufw status numbered
```

Step 6: Test Client-Server Communication

1. Start Exam Server on Ubuntu.
2. Run Exam Client from Kali.
3. Verify successful connection.

Expected Output:

Connected to Server

Question Paper Received

Step 7: Block Exam Port

Now block exam server port:

```
sudo ufw deny 5000
```

Check rules:

```
sudo ufw status
```

Step 8: Test Again from Client

Run client again.

Expected output:

java.net.ConnectException: Connection refused

Connection fails.

Firewall successfully blocks traffic.

Step 9: Allow Port Again

```
sudo ufw allow 5000
```

Test client again.

Connection restored.

Output:

Students should observe:

1. Firewall enabled successfully.
2. Port 5000 accessible when allowed.
3. Connection refused when port denied.
4. Connection restored after allowing rule.

Sample Output When Blocked:

Connection Failed!

```
java.net.ConnectException
```

Observations :

Action	Result
Default deny incoming	Blocks all external traffic
Allow port 5000	Client connects successfully
Deny port 5000	Client connection refused
Allow port again	Communication restored

8 Conclusion

This experiment successfully demonstrated the working of host-based firewall using UFW/GUFW.

It was observed that:

- Firewall filters traffic based on port rules.
- Default deny policy improves security posture.
- Unauthorized access can be effectively prevented.
- Firewall plays a critical role in protecting network services like the Online Examination System.

Thus, UFW/GUFW provides effective and simple firewall management in Linux environments.